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Norms prioritizing positive assessments are likely to maintain cooperation in private indirect reciprocity

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Exploring the evolutionary mechanisms of cooperation in societies where reputational consensus cannot be assumed, as in divided societies, is important for understanding the basic principles of human behavior in modern societies. Indirect reciprocity provides a major explanatory mechanism, but most studies have focused on how donors, i.e., one who decides whether to donate (cooperate or help) to a recipient or do nothing, are assessed. It is natural to think that there is no consensus among assessors in our society, and there is no reason to update only donor assessments when updating assessments. We constructed an agent-based model that enables updating of both donors' and recipients' images. Our exhaustive simulations showed that the well analyzed assessment rules updating donors' images are only second best to an assessment rule updating bad images in most likely for maintaining cooperative regimes. Such a social norm that prioritizes a positive assessment is considered tolerant, which is also consistent with previous studies arguing that tolerant evaluation is important in private-assessment schemes.

One of the most difficult questions in the context of the evolution of cooperation study is why cooperative behavior evolves between strangers. Cooperation between close relatives can be explained using the framework of kin selection[1] or selfish genes[2], and situations in which people interact with the same person over and over again can be explained using the principle of direct reciprocity or economic rationality that takes into account future value[1,3]. However, humankind has developed an advanced social system through solid cooperative relationships with strangers[4–6]. Indirect reciprocity is one of the few mechanisms to explain this[1,3,7–34]. Using indirect reciprocity, a cooperative society is maintained by eliminating uncooperative 'free riders' by cooperating only with strangers having good reputations. Thus, sharing information about who is a good person is a necessary condition for effective indirect reciprocity. In other words, society is required to implement public assessments and a reputation-management system to distribute them.

The situation in which social fragmentation or social division is accelerating, such as in modern society, poses new challenges to cooperative systems based on indirect reciprocity. In theoretical studies on the evolution of cooperation using indirect reciprocity, there has been a strong assumption that people must make assessments of others in accordance with the reputation information distributed by the reputation management system. However, public assessments or reputations are becoming ineffective in modern society[35,36]. This is because people's assessments of events or people greatly differ. Previous theoretical studies on indirect reciprocity, with the exception of a few studies[37–51], have not dealt with private assessment schemes in which the evaluations of others do not agree, so they have not provided a theoretical explanation of norms that function in a divided society. A new indirect reciprocity mechanism that is effective in our divided society needs to be determined.

Theoretical analyses on indirect reciprocity show that reputation sharing, which identifies potential partners as good or bad, is essential for solving social dilemmas. This simple policy, however, has an Achillesf heel[30]. When a someone adopting the above reputation-based action rule meets a bad person, that someone must take non-cooperative actions with that person. Such non-cooperation causes a second dilemma, i.e., the scoring dilemma[3], in which one's reputation lowers from a third party's point of view. The solution is not to base reputation decisions solely on behavioral information about cooperating or not cooperating but on complex assessment methods that

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allow for justified defection, i.e., consider on whom someone acts to. A number of theoretical studies have shown that assessment rules considering not only the first-order information of actions but also second-order information on whom someone acted are effective[3,13,15,17,18,23,34]. Despite theoretical analysis on complex assessment norms, there have been many experiments involving actual participants showing that people adopt simple assessment norms[52–56]. Therefore, a framework that matches theoretical insights and experimental results is needed.

Okada[46] pointed out that the theoretical idea of resolving the second dilemma with complex assessment norms in indirect reciprocity requires an extreme case (public assessment scheme) in which all participants in a society share all reputation information uniquely. If this scheme is not premised, it must be assumed that other people's assessments (hereafter, images) do not match, and even if we try to solve by conventional complex assessment norms, a third dilemma will occur, i.e., the punishment dilemma, in which even if punishment of free riders is encouraged by the social norms of a population, problems may arise if different individuals disagree on who deserves punishment. For example, if Chris thinks Bob is a good person and Alice does not cooperate with Bod because Alice thinks Bob is a bad person, then Chris does not correctly comprehend Alice's intentions and cannot justify them. Generally, when a society does not adopt a public assessment scheme, the justified-defection is not necessarily justified.

To make indirect reciprocity effective in a society that assumes a private assessment scheme, it is necessary to search for assessment norms that can resolve the above three dilemmas. Okada[46] exhaustively investigated all possible configurations of assessment rules and found two solutions to these dilemmas. However, almost all previous studies including the above have focused only on updating the assessments of the donor, i.e., one who decides whether to donate (cooperate or help) to a recipient or do nothing (defect). Few studies have considered the recipient's assessments[57]. Since the private assessment scheme assumes that the donor's and observer's assessments do not match, there is no reason to update only donor assessments. In the previous example, consider how Chris could update his assessment of Bob.

If Chris thinks that Bob is more trustworthy than Alice, Alice's defection against Bob would be a reason for Chris to update his assessment of Alice. However, if Chris thinks that Alice is more trustworthy than Bob, it may be Chris's assessment of Bob that should be updated upon observing Alice's defection. In other words, in the private assessment scheme, the object of assessment update should include not only the donor but also the recipient. However, since most studies assume public assessment schemes, this extension has been of little interest.

We conducted agent-based simulations to explore suitable norms for effective indirect reciprocity in a society with no single reputation for individuals. In other words, we exhaustively analyzed a private indirect reciprocity scheme in which updating not only the donor's assessment but also the recipient's one is permitted. While almost all previous studies considered only updating the donor's assessment when searching for appropriate assessment norms, in the case of private assessments, not only the donor's assessment but also the recipient's assessment should be considered.

Results

Agent-based model of social dilemma games

We construct an agent-based model here. Consider a well-mixed finite group of N individuals playing social dilemma games. In the game, there are two players: a donor and a recipient. The donor decide whether to donate (cooperate or help, denoted as C) or not (defect or do nothing, denoted as D) to recipients on the basis of their own strategies. As with many previous analytical studies, players have three strategies: to be the first-order free-rider who always chooses D (perfect defector, denoted as Y), the second-order free-rider (free-riding against the ability to eliminate the first-order free-rider) who always chooses C (perfect cooperators, denoted as X), and conditional cooperators adopting a given social norm (discriminator, denoted as Z). All players play two games at each time $t \in [1, T]$ where time is discrete. Among two games, each player participates in one game as a donor and in another game as a recipient. Every partner and the playing order are randomly determined at each time. Note that a total of $2NT$ games are played in all periods. In each game, When a donor chooses C, the donor pays the cost c while the recipient receives the benefit b, where $b > c > 0$. If the donor chooses D, nothing happens, that is, neither the donor nor recipient change their payoffs.

Even if donors choose C, C might not be executed with a small probability e_1; thus, D is executed instead of C. This is called an implementation error. The implementation error is valid for C only; thus, if donors choose D, D is executed. Each player in the population observes each game with a probability q. In other words, there is normally a mixture of individuals who observe the game and those who do not. There are also observation errors. Any observation content (C or D) is misunderstood with a small probability e_2; thus, the observer would recognize the opposite action for the actual one (either C or D).

An observer observing the game updates the assessment or impression (image) of the players privately on the basis of the observation result. Only discriminators (conditional cooperators) can become observers because they use players' images to decide whether C or D when they become donors. The image is binary, i.e., good ($= G$) or bad ($= B$). Note that public assessment schemes help make decisions (whether C or D) on the basis of the image (reputation) of the donor and that of the recipient, whereas in private assessment schemes the image (impression) is something they have privately. Thus, it is natural to make it a simple action function (C to G and D to B). In other words, the action rule of discriminators is choose C if a recipient's image in the mind of the discriminator is G; otherwise, choose D. Let an image that player i has on player j at time t be $A_{ij}(t) \in \{G, B\}$.

Definition of social norms

Observers of games have the opportunity to update their image of donors or recipients. How to update the player's image is an essential point. We introduce a consistency function to formulate this. It is a function for observers to determine whether a game being observed conforms to social norms. If an observer feels that a game

is consistent, they do not update the player's images (but may update the update time to current). However, if the observer feels that the game is not consistent, they update the player's images to be consistent. This consistency function is defined as

$$f_C : \text{Donor's image}\{G, B\} \times \text{Recipient's image}\{G, B\} \times \text{Donor's action to Recipient}\{C, D\} \to \{0, 1\} \quad (1)$$

where 1 represents "consistent" while 0 not. There are 256 consistency functions (because the total number of cases in the domain is $2 \times 2 \times 2 = 8$, and there are 2^8 ways to assign either 0 or 1 to each case) and numbered in the Supplementary Information (SI). In each simulation run, one consistency function is fixed given and adopted as a social norm. We did not consider any case of multiple consistency functions competing against each other in the simulations.

If an observer sees a game as inconsistent with social norms, how do they update a player's image? Almost all studies to date dealt only with updating donor images. However, this constraint is too strong for private assessment schemes. Here, we consider the possibility updating the images of recipients as well as donors. We consider the following six rules, which include the most common rule (Rule 1).

[Rule 1: Prioritizing Recipient's Image] Prioritize the recipient's image and update the donor's image
[Rule 2: Prioritizing Donor's Image] Prioritize the donor's image and update the recipient's image
[Rule 3: Prioritizing New Image] Prioritize the most recently updated image and update the older image
[Rule 4: Prioritizing Good Image] Prioritize good images and update bad images
[Rule 5: Prioritizing Bad Image] Prioritize bad images and update good images
[Rule 6: Random Updating] Randomly decide what image to update

In Rules 3–5, the donor and recipient images may have the same priority; thus, if so, one image (either donor or recipient) is selected at random for updating.

The total number of possible social norms in this model is $256 \times 6 = 1536$, where each norm follows one single rule. For example, a social norm called [f_C^{150}, Rule 4] means that the 150th consistency function under [Rule 4: Prioritizing Good Image] as the image update rule is adopted. We comprehensively analyzed all possible social norms.

Simulation results for all social norms

Our agent-based simulation results for all social norms indicate that, on the one hand, Rules 2, 3, 5, and 6 cannot maintain cooperation of any consistency function except for 231 (Fig. 1). On the other hand, Rule 1 (hereafter referred to as Donor Updating Rule) and Rule 4 (hereafter referred to as Good Image Prioritizing Rule) have several consistency functions that can maintain cooperation. In other words, what is prioritized when updating the image is an important factor as a social norm for maintaining a cooperative regime. It is interesting to note that in contrast to the Donor Updating Rule, Rule 2 (which favors the donor image and updates the recipient image) fails to maintain cooperation with any consistency function. It might make sense that Rule 6 (Random Updating) does not work, but surprisingly Rule 3 (Prioritizing New Image) does not work either. Rule 5, the rule to update good images in favor of bad images, failed to maintain cooperation for any consistency function.

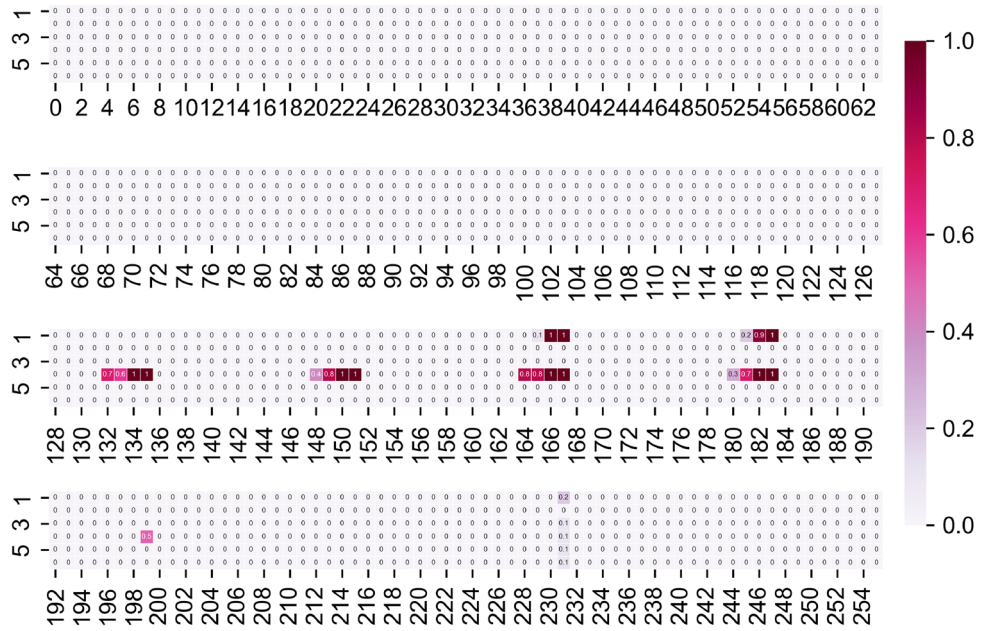


Figure 1. Rate of maintaining cooperation in all possible social norms. This heat map shows percentage of simulation runs where cooperation rate exceeded 80% at end of 100th generation for each social norm [f_C^x, Rule y], where horizontal axis represents x and vertical axis represents y. Each social norm was run 10 times with different random seeds. Parameter values were $N = 100$, $(b, c) = (3, 1)$, $(e_1, e_2) = (1\%, 1\%)$ and $q = 0.1$.

In contrast, the Good Image Prioritizing Rule, the rule to update bad images in favor of good images, was able to maintain cooperation with various consistency functions. These results reveal the necessary conditions for priority rules that can sustain cooperation for several consistency functions. It should be either a rule that prioritizes the recipient image and updates the donor image, or a rule that prioritizes the good image and updates the bad image.

Stable cooperative norms determined

As the exhaustive results shown in Fig. 1 reveal, only a few social norms can sustain cooperation. We then show in Fig. 2 the results of examining the distribution of the cooperation rate for the social norms where the rate of maintaining cooperation exceeds 0 in Fig. 1. Out of the distribution of cooperation rates in the box plot of Fig. 2, a social norm in which the cooperation rate at the third quartile exceeds 80% is called a stable cooperative norm (SCN). As shown in this figure, the SCNs under the Donor Updating Rule (Rule 1) are social norms with consistency functions are 166, 167, 182, or 183. SCNs under the Good Image Prioritizing Rule (Rule 4) are those with consistency functions 134, 135, 150, 151, 165, 166, 167, 181, 182, or 183. As can be seen in Fig. 2, there are other social norms that is possible to maintain cooperative systems, but they are not stable due to the effects of randomness of the simulations. Note that social norms that can be stabilized by the Donor Updating Rule can also be stabilized by the Good Image Prioritizing Rule. Therefore, it is suggested that the Good Image Prioritizing Rule is the easiest rule for constructing SCNs.

Features of SCNs in donor updating rule

Next, we investigated what consistency functions constitute the SCNs. We first consider the Donor Updating Rule. This rule has been typically assumed in almost all previous studies on indirect reciprocity. Ohtsuki and Iwasa[13] identified eight norms that can sustain cooperation in public assessment schemes. These norms are known as the Leading Eight, which correspond to consistency functions 148–151 and 180–183. Consistency functions 180–183, called the Leading Four, can maintain cooperative regimes even in private assessment schemes[46]. In contrast with those studies, our simulations allow only two norms, called L4 (or consistency function 182) and L7 (or Staying, consistency function 183) to be included in SCNs. Why are our simulations more intolerant? This is because our simulations are conducted in finite populations not a mathematical analysis in infinite populations, and we assume that the effects of perturbation originating from mutation and the learning process is not small.

In the Donor Updating Rule, two norms (consistency functions 166 and 167) are included in SCNs. These norms have a remarkable feature: they can keep a cooperative regime both in private assessment and in public assessment schemes, but stable populations consist of a mixture of norm-adopters and perfect cooperators, as shown in the SI.

Image scoring maintain cooperation in good image prioritizing rule

We find 10 SCNs in Good Image Prioritizing Rule of all 256 consistency functions. The most important SCN under the Good Image Prioritizing Rule may be consistency function 165. The social norm called $[f_C^{165},\ \text{Rule 1}]$ corresponds to Image Scoring, the simplest strategy for indirect reciprocity. The image-updating pattern of Image Scoring is shown in Fig. 3a. As many theoretical analyses including that by Sigmund[3] point out, Image Scoring

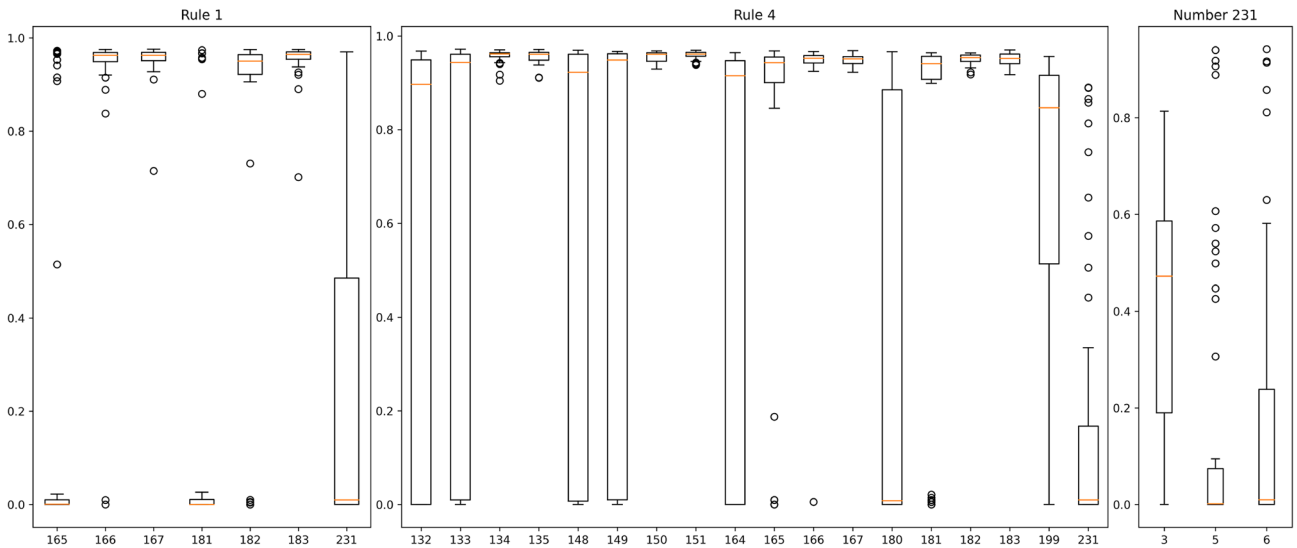


Figure 2. Cooperation rates in each social norm. This box plot shows cooperation rates at end of 100th generation for each social norm. Each social norm was run 50 times with different random seeds. Left panel represents results of Donor Updating Rule (Rule 1) and center panel represents results of Good Image Prioritizing Rule (Rule 4), where horizontal axis represents x of f_C^x consistency function. Right panel represents results of consistency function 231 where horizontal axis represents y of $[f_C^{231},\ \text{Rule } y]$. Parameter values were as in Fig. 1.

temporally reaches a cooperative regime, but cooperation often breaks down in very long-term simulations. In our simulations, however, $[f_C^{165}$, Rule 4] shown in Fig. 3b maintained cooperative regimes. Compared with the normal Image Scoring (Donor Updating Rule), this norm (Good Image Prioritizing Rule) differs in three cases of the updating rule, as shown in Fig. 3a, b. This feature may re-shed light on Image Scoring.

Norms adopting Heider's balance theory maintain cooperation

The social norm $[f_C^{150}$, Rule 1] (or Stern-judging or Kandori) included in Leading Eight cannot maintain cooperation in the private assessment scheme, but social norm $[f_C^{150}$, Rule 4] (Fig. 3c) under the Good Image Prioritizing Rule is included in SCNs. This norm corresponds to Heider's balance theory[58]. Thus, the interpretation of this norm is updating an image of either the donor or recipient as prioritizing good images if the game in the eyes of an observer is inconsistent with respect to the view point of Heider's balance theory for the triplet of donor's image, recipient's image, and donor's action.

Figure 4 shows a summary of known cooperation norms and SCNs. There are social norms that have similar characteristics and have close consistency function numbers; thus, we group some norms together. The Heider type (consistency function 150) has assessment rules following the Heider's balance theory for all eight cases of a game. Let us consider another social norm that does not follow the theory for only at most two of the eight cases: defecting to bad recipients by either good and bad donors. These cases have often been analyzed as so-called justified defection. Let us extend the Heider type of allowing assessments for such a justified defection to be freely made and call such extended social norms Heider-like norms. As shown in Fig. 4, consistency functions 134, 135 and 151, which are Heider-like norms, are SCNs under the Good Image Prioritizing Rule. We call Image-Scoring-like norms those where assessments for a game played by bad donors and bad recipients are made freely while the other assessments follow like Image-Scoring norm. Three Image-Scoring-like norms (consistency functions 165, 166, and 167) are SCNs under the Good Image Updating Rule.

Although we defined SCN above, there are many ways to define it. Despite such flexibility, the most stable social norms may be narrowed down to several candidates. In our simulation results, $[f_C^{167}$, Rule 4] (Fig. 3d) and $[f_C^{183}$, Rule 4] are included as strong candidates for the champion which can maintain cooperative regimes most robustly. Consistency Function 183 corresponds to Staying (L7), and consistency function 167 has characteristics like mixing of Image Scoring (consistency function 165) and Staying. See details in the SI.

Discussion

Exploring the evolutionary mechanisms of cooperation in societies where reputational consensus cannot be expected, as assumed in divided societies, is important for understanding the basic principles of human behavior in modern societies. Although indirect reciprocity provides a major explanatory mechanism, only a few studies have analyzed the evolutionary mechanisms in a private assessment scheme where reputational consensus is not guaranteed. However, almost all studies on private assessment schemes have focused on how donors' images are assessed despite there is no foundation to consider only updating the images of donors in a private assessment

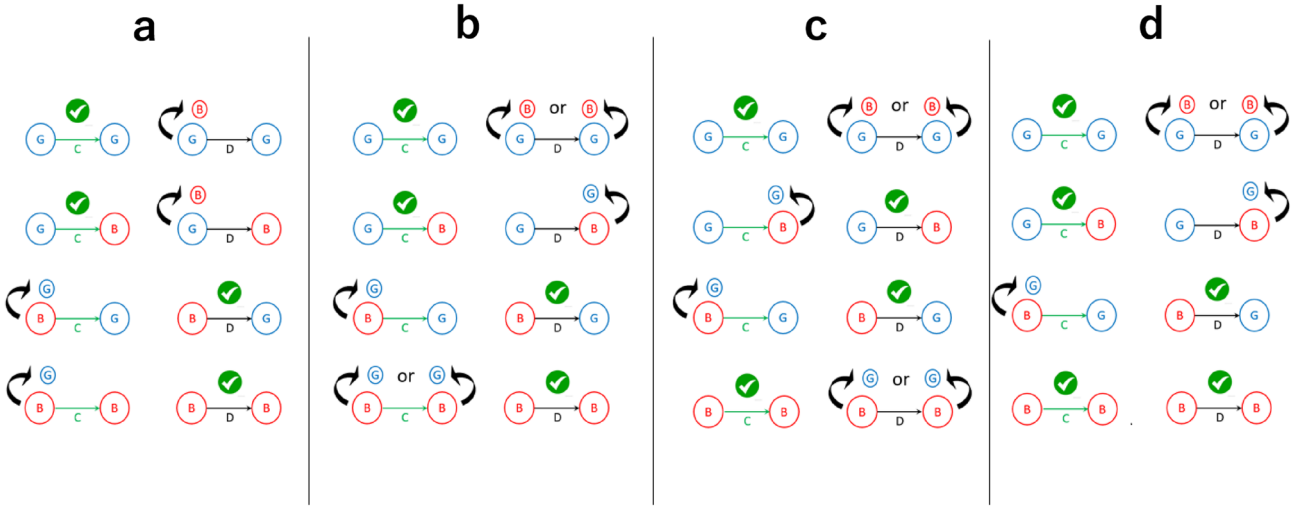


Figure 3. Configuration of image-updating patterns in social norm $[f_C^x, Ruley]$: **(a)** $(x, y) = (165, 1)$, **(b)** $(x, y) = (165, 4)$, **(c)** $(x, y) = (150, 4)$, **(d)** $(x, y) = (167, 4)$. Each panel shows image-updating patterns for all eight cases corresponding to the domain in the definition equation of the consistency function (Eq. 1). In each case, donor's image is either *G* or *B* in left circle, recipient' image is either *G* or *B* in right circle, and donor's action is either *C* or *D* in center arrow. If observer observes one of eight cases, observer recognizes whether case is consistent. If it looks consistent (with green check mark), no image will be updated. If not, either donor's image or recipient's image will be updated in accordance with diagram using curved arrows and small circles in each case. Note that (**a**) corresponds to normal image scoring while (**b**) represents Image Scoring under the Good Image Prioritizing Rule, and (**c**) represents L6 (or Kandori, Heider-type) under the good image prioritizing rule.

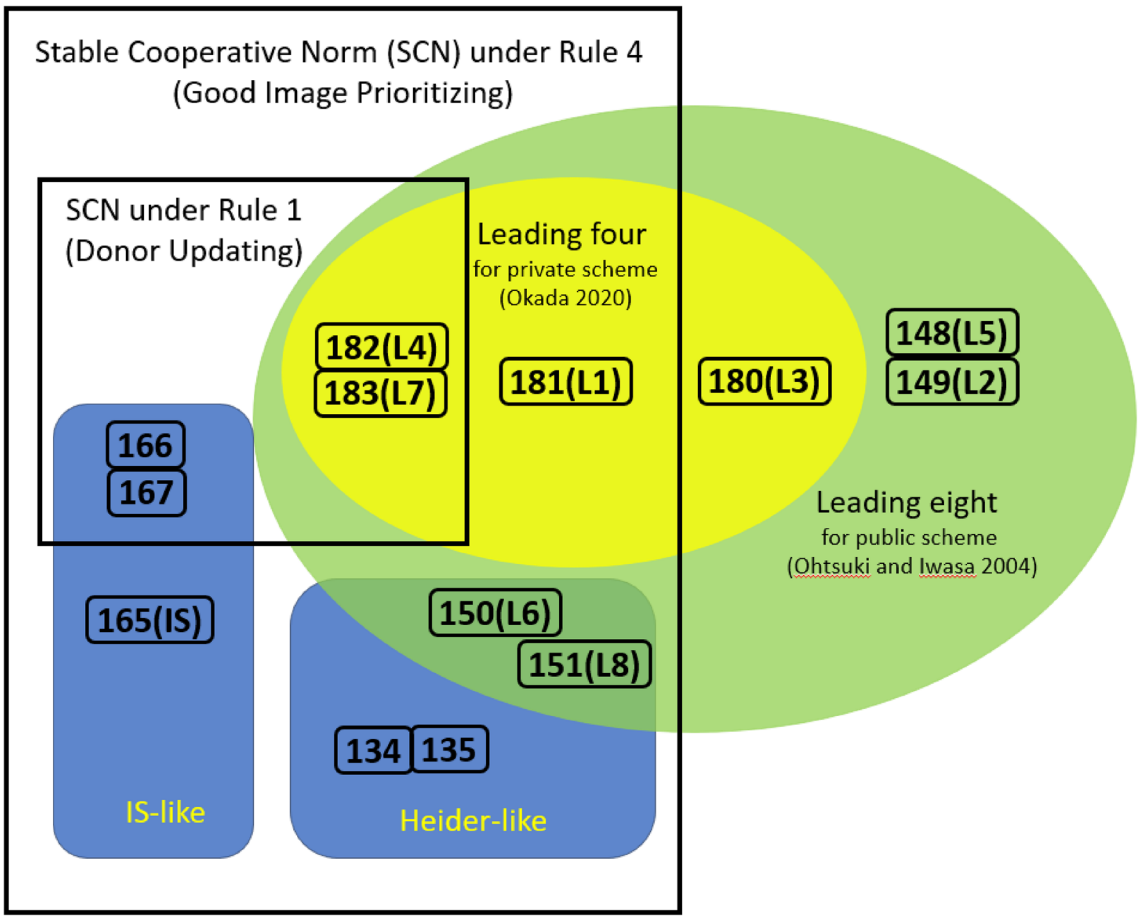


Figure 4. Summary diagram of relationship between SCNs and known cooperation norms. This diagram shows (1) SCNs under Donor Updating Rule, (2) SCNs under Good Image Prioritizing Rule, (3) Leading Eight discovered by Ohtsuki and Iwasa (2004), (4) Leading Four discovered by Okada (2020), (5) Image-Scoring norm and its similar norms (IS-like), and (6) norm consistent with Heider's balance theory and its similar norms (Heider-like). Diagram number corresponds to consistency function number while number beginning with L in blanket corresponds to number of Leading Eight given by Sigmund (2010). Note that although this diagram shows the SCNs and the Leading Eight correspondences, this correspondence holds "only in the Donor Updating Rule (Rule 1)" and not in Rule 4.

scheme. We, therefore, constructed an agent-based model that enables updating the images of both donors and recipients. Our exhaustive simulations showed that the well-analyzed assessment rules, which are updating donors' images, are only the second best to an assessment rule updating bad images for most likely maintaining cooperative regimes. Specifically, when it is necessary to update the image of either a donor or a recipient, it is desirable to adopt an assessment-updating rule that changes the image of a person with a bad reputation, regardless of whether one is the donor or recipient. Such a social norm that prioritizes good images is considered a tolerant norm and consistent with previous studies arguing that tolerant evaluation is important in private assessment schemes[34,41,43,46,59].

In this paragraph, we infer why Good Image Prioritizing rules be more successful than Donor Updating rules. These rules correct bad assessments regardless of the donor or recipient, so it is a tolerant assessment rule. Specifically, when it is necessary to update the image of either a donor or recipient, it is desirable to adopt an assessment-updating rule that changes the image of a person with a bad reputation, regardless of whether one is the donor or recipient. Many studies[41–43,45–51] have shown that the rules under which indirect reciprocity works in private assessment schemes are more limited than in public assessment schemes. Such a social norm that prioritizes good images is considered a tolerant norm and consistent with previous studies arguing that tolerant evaluation is important in private assessment schemes[34,41,43,46,59]. On the basis of this study's original model, which includes the recipient's image as a subject of updating, we argue that it is possible to maintain cooperative regimes with more norms than have been discovered thus far.

The main contribution of this study was to find social norms that maintain cooperation. In particular, the Good Image Prioritizing Rule is easier to maintain than the Donor Updating Rule, which has been extensively analyzed in previous studies, and has led to the theoretical discovery of several new social norms. We should mention two important social norms that are stable cooperative norms under the Good Image Prioritizing Rule. The first is Image Scoring. This norm discovered by Nowak and Sigmund[8] is theoretically known as the simplest social norm for explaining indirect reciprocity. This is as simple as assessing cooperators as good and

non-cooperators as bad (Fig. 3a). However, many theoretical studies repeatedly showed that this norm has not been completely stable (called the scoring dilemma or the second dilemma), and they argue that humans may adopt more complicated assessment norms. However, several experimental studies did not observe such a complex social norm, which has caused a long debate between theoreticians and experimental researchers. We found that Image Scoring under the Good Image Prioritizing Rule (Fig. 3b) is possible to maintain cooperation regimes. Image Scoring may have features we do not know about yet.

The second norm is the so-called Stern-judging (or Kandori, L6) norm[60]. This norm had mixed reviews among experts. Some theorists argue that its assessment rules are too strict to be a stable cooperative norm in private evaluation schemes, while others argue that only social norms having such a strict assessment rule maintain cooperative regimes in low frequency of learning[19,45,61]. However, a judgement of whether it is consistent or inconsistent in all considerable cases of a game is the same as the Heider's balance theory. This theory empirically supported by many studies expresses a human's tendency to maintain the relationship of the assessments (either positive or negative) of three objects so as to be cognitively consistent as one of the cognitive consistency theories[62]. The Stern-judging norm has an assessment rule that applies the three-object relationship in the balance theory to the image of the donor, image of the recipient, and action of the donor to the recipient. Our simulations showed that this norm under the Good Image Prioritizing Rule (Fig. 3c) maintains robust cooperation even in private assessment schemes. Our results suggest that Heider's balance theory works to drive indirect reciprocity by adopting the Good Image Prioritizing Rule.

Through our analysis, we were able to confirm the superiority of the Donor Updating Rule than Recipient Updating Rule, which has no social norm to maintain cooperative regimes. It is not clear what the source of the asymmetry in such updating is. The domain of the consistency functions in our model consists of one type of information derived from the recipient and two types of information derived from the donor. This point may be related to such asymmetry, but future study is required. It is also surprising that the Prioritizing New Image Rule (Rule 3) does not work. It is natural to target older information when we have to fix an inconsistent set of images. However, our simulations showed that it does not work at all. This point should also be for future work.

It should be noted that there are several limitations in the current version of our study. Our model handles only a restricted case of updating either a donor or recipient image for each observer. However, there are theoretical cases of updating both images. We used a coin-toss type with respect to updating either donor or recipient simply when two images are equally chosen as alternatives. The current version of this paper is the first step to analyzing the extended model which does not have the limitation that an updating object is only a donor image; thus, we prefer the simplest rule. However, there might by significant results when loosening the above assumptions. Another extension may be to consider the impact of different network structures on the emergence of cooperation. Social networks in the real world are rarely well-mixed, and incorporating this complexity could reveal important dynamics. In the model, we assumed the six rules with respect to image updating. Although we need empirical evidence that support these assumptions, there has been few studies in social psychology regarding this. Future study should include an experiment that involves using a framework that allows changes in the images of both donor and recipient or uses settings such as game observation that assumes Heider's balance theory.

Methods

We conducted agent-based simulations to clarify whether each social norm can maintain a cooperative regime in a society where a certain social norm is adopted. The strategies of players (X, Y, Z) are updated by the Fermi learning rule expressing that the higher the player's payoff[63], the more advantageous the player's strategy. We call the T periods one generation, and let the expected payoff per play earned by Player i in one's last $T'(<T)$ periods at the end of each generation be one's fitness value Π_i. Note that the number to be selected as a donor (and as a recipient) in each periods is T'. That is, using the payoff $\pi_i(t)$ for player i at time t,

$$\Pi_i = \frac{1}{T'} \sum_{T-T'<t\leq T} \pi_i(t). \tag{2}$$

At the end of each generation, every player compares one's fitness value with a randomly chosen player (j) and uses the Fermi-type learning rule[63] to determine player i's strategy for the next generation. Specifically, player i changes to player j's strategy with a probability $p(i \to j)$ where

$$p(i \to j) = \frac{1}{1 + e^{(-\beta(\Pi_j - \Pi_i))}}. \tag{3}$$

However, regardless of whether player i changes their strategy to player j's, it is assumed that player i will be replaced with a mutant with a probability μ. If replaced by a mutant, nature chooses one of the three strategies (X, Y, Z) at random. In our simulations, we set the initial population distribution of the strategy as $(\#X, \#Y, \#Z) = (1, 1, N-2)$ and all images as G, i.e. $A_{ij}(t=0) = G$. The other parameter values were $(T, T') = (1000, 100)$, $\beta = 3$ and $\mu = 1\%$. Results of robustness checks on key parameters of the model are presented in the Supplementary Information.

Data availability

We declear that the data utilized in this study were obtained independently of human involvement. The datasets generated and analyzed during the current study are available in the Dataverse repository, https://doi.org/10.7910/DVN/B3KUYB.

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Author contributions

All authors conceived the study, discussed the results, and wrote the manuscript. I.O. developed the model, and analyzed the results.

Competing interests

The authors declare no competing interests.

Additional information

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